

A new operational profile, preserving v0.7

Authorial corpus: Tom Klootwijk. Identifier: NL200678942. Date as supplied: 10-07-1990. Release: TOM v0.7.1. This operational extension was specified on 19 September 2026 following the author's instruction to add executable behavior. The identifying details record authorship; they are not physical constants, encryption keys or simulation parameters.

v0.7.1 retains the whole v0.7 definition and assigns one explicit continuation profile to previously open behavior. The chosen Thue-Morse jitter and exact golden-ratio recurrence are new model assignments. They are not claimed to follow from v0.7, to have been present in the original corpus, or to predict physical events.

TOM := InverseOccam[Arch[C[J[T]]; E[C[J[T]]; InvLog[phi]]] qualified_by forward_T_precedence

The abbreviations are $c := C[J[T]]$ and $e := E[c; \text{InvLog}[\phi]]$. Semicolons distinguish ordered roles. Expansion changes spelling, not the definition. SDF continues to mean the native admission qualification: it does not assert a metric or a normalized signed-distance function.

What this release makes definite

A committed state supplies exact coefficients a and b . At the next admitted simulation tick, the profile computes a deterministic jitter sign and derives a new state. It then constructs the exact echo, retains both ordered roles, records the zero scalar reading without replacing the definition, and commits the complete step only if time, evidence and capacity checks pass.

In plain language: the earlier rulebook said how to preserve the past and wait for a valid next step. This version also specifies one rule that calculates that next step. It is a repeatable mathematical process with an inspectable history; it is not a model of every phenomenon that might later be described using TOM.

Governing source and status

The unchanged parent is TOM_V0_7_Formalization.pdf (18 pages), SHA-256 97582a16bd54d690b27105f9b09dc48d2b5716bbae2fff2a5e74190278640fbc. Its pages 3-9 define the binding and structural contracts; page 14 identifies open assignments. The original PDF and package are retained. This document is a specification, not a claim that acceptance tests or a performance benchmark have passed. Execution evidence belongs in a separately dated results report.

The structural commitments remain in force

Every admitted defining role, compound and admissibility rule remains SDF-qualified. T, J, lowercase phi, InvLog, C, E, Arch and InverseOccam retain their roles. The profile imposes no intrinsic coordinate grid, vector space, operator graph, quaternion arena or global Psi enclosure. Computer records and their operand addresses are a disclosed external representation.

Commitment	Requirement
TC1 / no anticipation	Use only supplied initial evidence available at or before the committed simulation time. Deriving a simulated future is not possessing an observation of a physical future.
TC2 / completed prefix	Never alter a committed record or its order. A correction is a later record or a separately declared new experiment.
TC3 / no time reversal	An inverse-log role or a negative coefficient does not reverse the temporal qualification.
TC4 / guarded self-reference	Own-name declaration is permitted. The result of continuation becomes available only after the next step is derived and admitted.
TC5 / elapsed time	Each admitted step advances by its declared positive delta. Nesting or a zero readout does not remove that elapsed duration.

Only licensed equivalences

InverseOccam[InverseOccam[x]] =def InverseOccam[x]

The other licensed representational changes are c/e/TOM abbreviation expansion and A =def T only when explicitly declared. A single InverseOccam wrapper is not identity. Ordered Arch and E operands cannot be swapped, flattened or reassociated by appealing to a symmetric numeric reading. There is no implicit InvLog involution.

Equality of selected readings does not imply equality of definitions. Normalization retains the original source and original terms. A finite representation may share equivalent storage, but it must not erase the defining distinction or its provenance.

There is no fixed intrinsic nesting depth. The executable has declared storage and address bounds; failing to fit is a realization failure, not a new source law. Klein and Matryoshka retain their named roles. This release assigns neither a Klein topology nor a nested physical construction to those names.

Exact numbers, jitter and the next state

$\phi = (1 + \sqrt{5})/2$; $\phi^2 = \phi + 1$; $\kappa = -\ln(\phi) < 0$

$C_n = a_n + b_n \cdot \phi$, with a_n, b_n in the integers

The default seed is $(a_0, b_0) = (0, 0)$, with $n=0$ and $\text{time}=0$. A caller may explicitly supply any representable signed integer pair and nonnegative step index. Such a seed defines the start of that experiment; it does not assert that an earlier history has been supplied. An arbitrary seed is not interchangeable with resuming an existing saved experiment.

$J_n = (-1)^{\text{popcount}(n)}$

$J_0 = +1$; $J_{(2n)} = J_n$; $J_{(2n+1)} = -J_n$

This is the signed Thue-Morse sequence, selected as a deterministic jitter law. It is not random noise, a measured frequency, thermodynamic entropy or a security primitive.

$C_{(n+1)} = \phi \cdot C_n + J_n$

$(a_{(n+1)}, b_{(n+1)}) = (b_n + J_n, a_n + b_n)$

The coefficient update follows by substituting $\phi^2 = \phi + 1$. Computational states use signed integer limbs and exact symbols only: no floating values, approximate square roots, logarithms or divisions enter the transition. The irrationality of ϕ makes (a, b) a unique representation of $a + b \cdot \phi$.

input n	J_n	before (a,b)	after (a,b)
0	1	(0, 0)	(1, 0)
1	-1	(1, 0)	(-1, 1)
2	-1	(-1, 1)	(0, 0)
3	1	(0, 0)	(1, 0)
4	-1	(1, 0)	(-1, 1)
5	1	(-1, 1)	(2, 0)
6	1	(2, 0)	(1, 2)
7	-1	(1, 2)	(1, 3)
8	-1	(1, 3)	(2, 4)
9	1	(2, 4)	(5, 6)
10	1	(5, 6)	(7, 11)
11	-1	(7, 11)	(10, 18)

These twelve values are mathematical examples derived from the stated recurrence. They are acceptance vectors, not evidence of a passed executable test. With $\delta=1$, each row commits at time $n+1$. Every row retains its own history even when a later coefficient pair repeats an earlier pair.

The whole expression has an operational reading

Role	Selected v0.7.1 meaning
T	The current committed index n, simulation time t, positive delta and supplied horizon H.
J[T]	Compute the exact sign J_n from the current n.
C[J[T]]	Use the committed coefficient pair and the evaluated jitter operand to derive (b+J_n, a+b).
InvLog[phi]	The exact named constant $\kappa = -\ln(\phi)$. Its representation is symbolic, not a rounded decimal.
E[C; InvLog[phi]]	Construct the exact real expression $\kappa * (a_{\text{next}} + b_{\text{next}} * \phi)$ from its actual evaluated operands.
Arch[C; E]	Retain the ordered direct/echo pair and calculate the selected external scalar reading as exactly zero.
InverseOccam[...]	Preserve the defining expression, the original supplied source and the complete committed records.
forward T qualification	Admit and publish a step only after the transition and all temporal/capacity obligations succeed.

$E_{(n+1)} = \kappa * (a_{(n+1)} + b_{(n+1)} * \phi)$
 $\text{arch}(x, y) = |x| + |y| - |x - y|$
 $\text{arch}(C_{(n+1)}, E_{(n+1)}) = 0$

The zero result is exact: κ is negative and the arguments have opposite signs or are zero. The echo stores the derived integer coefficients together with the exact κ tag. There is no loss of precision merely because a transcendental constant is kept as a defined symbol. This profile does not evaluate arbitrary expressions involving transcendental constants.

Execution must dispatch through the actual admitted term records and operand references. It must reject a changed program that is no longer the full source expression under licensed equivalences. Running an unrelated recurrence beside a decorative copy of the expression does not satisfy this contract.

SELF and Later_T

The continuation driver uses SELF to refer to the actual committed coefficient state. Later_T is realized by the next admitted transition. Its result is the derived coefficient pair and preserved echo/Arch record; the caller never supplies a next-result coefficient pair as a substitute for computation. This profile admits only the complete TOM root, not arbitrary alternate programs or unrestricted recursion.

A tick is an atomic, bounded transition

A request supplies an initial or resumed state (n,t,a,b) , $\delta > 0$, a simulation horizon H , the latest available initial-evidence time, and a nonnegative requested step count. The same δ applies throughout that run. The supplied evidence time must not exceed t . The initial t must not exceed H . The units of these tags are explicit experiment parameters; no physical clock is authenticated.

Phase	Exact behavior
Validate	Reject malformed encodings, unsupported program shape, negative zero, zero δ or an initial time beyond the horizon. Reject future initial evidence.
Check next tick	First require room for a complete record; then require $n+1$ and $t+\delta$ to fit their 64-bit fields; then require $t+\delta \leq H$. A failed check leaves the current head unchanged.
Derive	Evaluate J_n and the full program; compute exact candidate coefficients, echo and ordered Arch result in disposable scratch.
Check storage	Require all candidate coefficients and the entire next history record to fit the declared capacities. Never wrap, truncate, erase, silently resize or substitute a floating result.
Commit	Publish one complete record and update (n,t,a,b) together. Then proceed to the next requested step.

Atomicity is per step. If a request commits three steps and the fourth fails, the first three remain committed. The failing step publishes no partial record and makes no change to the committed head. A request for more records than S can retain commits up to S successful steps and then reports history capacity, unless an earlier obligation stops it.

Status bit	Meaning
1 / INVALID	Invalid seed, encoding or temporal input.
2 / PENDING_HORIZON	The next positive tick lies beyond the supplied horizon.
4 / COEFFICIENT_CAPACITY	An exact candidate coefficient exceeds the selected limb capacity.
8 / TIME_CAPACITY	The next index or simulation time cannot fit its 64-bit field.
16 / HISTORY_CAPACITY	The next complete record cannot fit.
32 / PROGRAM_ERROR	The admitted-program execution contract fails.
64 / FUTURE_EVIDENCE	Initial evidence is tagged later than the initial committed time.

Initial validation can combine INVALID and FUTURE_EVIDENCE. An admitted run stops at its first failed check, in the order above; program evaluation follows retained term order. A zero status means the requested steps completed. Counts describe the successful prefix. Benchmark repeats restart the same seed; they are not additional evolution. A new run may explicitly change δ or horizon without rewriting its prior head.

Exact storage, observation and resume

The executable codec is part of this external profile. Words are little-endian uint32. A coefficient has a sign (0 positive/zero, 1 negative) and L little-endian base-2³² magnitude limbs. Zero must have sign 0. Its magnitude is at most 2^{^(32L)-1}. Index, time, delta, horizon and evidence tags are unsigned 64-bit values.

The source is retained as exact UTF-8 bytes, with zero padding only to the next word boundary. Original eight-word native term records and the original root are retained. Normalization licenses are the v0.7 licenses, not a new rewrite system. The Python source compiler checks source-to-term binding; a C++ host that retains arbitrary source text does not thereby parse or authenticate that text.

Quantity	Size in uint32 words
Seed case	16 + 2L
Committed record	16 + 6L
Disposable scratch per case	N*(4 + 2L)
Result case	48 + 4L + S*(16 + 6L)

N is the retained term count and S the next-run record capacity. Host size arithmetic must be checked before allocation. The file envelope has 16 words: magic, ABI version, case count, L, S, N, original root, flags, case words, result words, record words, UTF-8 byte count, archived prior-file words, then three reserved zeros. Input magic is 0x314f4d54 and result magic is 0x31524d54. ABI version is 1. The only program flag is bit 0 for explicit A=T.

After the envelope come the padded source bytes, original terms, the complete archived prior output (if any), and current cases. Fresh input files have no archive. A resumed output nests the entire previous output byte for byte, including its previous archives. Complete prior content is retained, not replaced by an unchecked hash or a count.

Each committed record stores input n, before/after time, jitter sign, before/after coefficient signs and magnitudes, exact echo coefficients and the kappa tag, scalar zero readout and qualification flags. These are simulation observations of the defined process. They do not acquire a physical interpretation merely by being called observations.

Before resume, the host re-evaluates every archived seed using its retained program and compares every saved result word. It also checks each segment starts from the preceding terminal coefficient pair, n and t, case by case. Limb capacity may grow without changing the integer value; it may not shrink. The new input must match the latest saved head. A zero-commit failure leaves that head equal to the original seed.

Buffers are reserved before execution. Output paths may not alias inputs or prior/reference artifacts. Checked temporary writes and atomic file replacement preserve an existing destination on publication failure. Multiple files are not one atomic transaction. There is no power-loss durability, cryptographic authorship or adversarial tamper-authentication claim; consistently rewriting an unauthenticated experiment cannot in general be detected from its own contents.

Consequences, acceptance and the boundary of this release

Consequences of the selected laws

Determinism: fixed seed, program, parameters and capacities give the same committed prefix and stop condition. Exact closure: multiplication by phi maps the integer pair (a,b) to (b,a+b), so adding J keeps the state in $Z[\phi]$. Strict progress: each successful transition adds one to n and a positive delta to time. Preservation: each commit appends a complete record; rejection leaves the prior head. Distinct histories remain distinct even when their final coefficient pairs or scalar zero readings agree.

These are consequences of the specification, not a formal verification certificate for a particular binary. Exactness holds within the announced coefficient/address capacities; exhaustion is reported rather than approximated. The default recurrence may revisit a coefficient pair while its index, time and retained history continue forward.

Required executable evidence

Validation must compare all result words with an independently implemented integer reference, including the first twelve steps; test changed full-expression bindings, licensed aliases and idempotence; exercise large positive and negative seeds; and test coefficient, clock, horizon and history boundaries. Resume must agree with uninterrupted evolution where parameters agree. Corrupt saved states, traces and inconsistent source/program bindings must be rejected within the stated parser/trust scope. Output collision and failed-publication checks must preserve existing files.

A performance report must name the workload, number of evolved steps, exact capacities, memory use, timing scope and verification method. Filling GPU memory is not proof of peak arithmetic utilization. A measured result must not be advertised as exhaustive exploration of every TOM expression or as a forecast of real events.

Assigned here; still open beyond this profile

Assigned here: deterministic jitter, an exact coefficient recurrence, a symbolic inverse-log echo, step/horizon actualization, a simulation observation record and a recoverable finite archive codec. The inverse-log and entropic names do not become measured physical mechanisms. No thermodynamic entropy, physical measurement protocol, general intrinsic pinion dynamics, Klein topology, autonomous discovery mechanism or arbitrary-SDF execution semantics are supplied.

A later profile may define different dynamics while preserving the v0.7 commitments. Such an extension must state its added law, its interpretation and its own validation. v0.7.1 is fully specified for this particular operational profile; it does not claim that every possible application or physical meaning of the corpus has been implemented.

Companion records

TOM_V0_7_1_Formalization.json is the machine-readable declaration accompanying this document. The kernel project's docs/V071_ABI_DESIGN.md gives the complete binary field layout and implementation agreement. The original v0.7 PDF remains the authority for unchanged clauses; this release is the authority for the explicitly selected new operational assignments.